

Engineering Approach: Deal Evidence Ledger

Evidence-native CRM · first product · MVP architecture, team and implementation plan
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AI interprets. Policy computes. People decide.

Six pages for a quick read. The detailed appendix is `ENGINEERING_APPROACH.md` (same system, same entities, more depth). Figures marked **[proposed]** are for agreement with engineering and a design partner; nothing is validated with customers yet.

1. MVP and engineering principles

Goal. Before a forecast call, identify the important deal claims that are **unsupported, contradicted or stale**, with the **exact evidence** for each. Today a CRM stores `FIELD = VALUE`; this product stores `CLAIM + EVIDENCE + SOURCE + TIME + HISTORY + STATUS`, and fields become views of it.

The principle: AI interprets. Policy computes. People decide. Policy computes five statuses: **SUPPORTED · UNSUPPORTED · CONTRADICTED · STALE · UNKNOWN**. A human decision is stored beside the status and never rewrites it.

Users. Front-line **sales manager** (primary) · **account executive** (secondary) · **CRO / VP Sales** (economic buyer) · **RevOps** (operational champion). Workflow: the **weekly forecast review**.

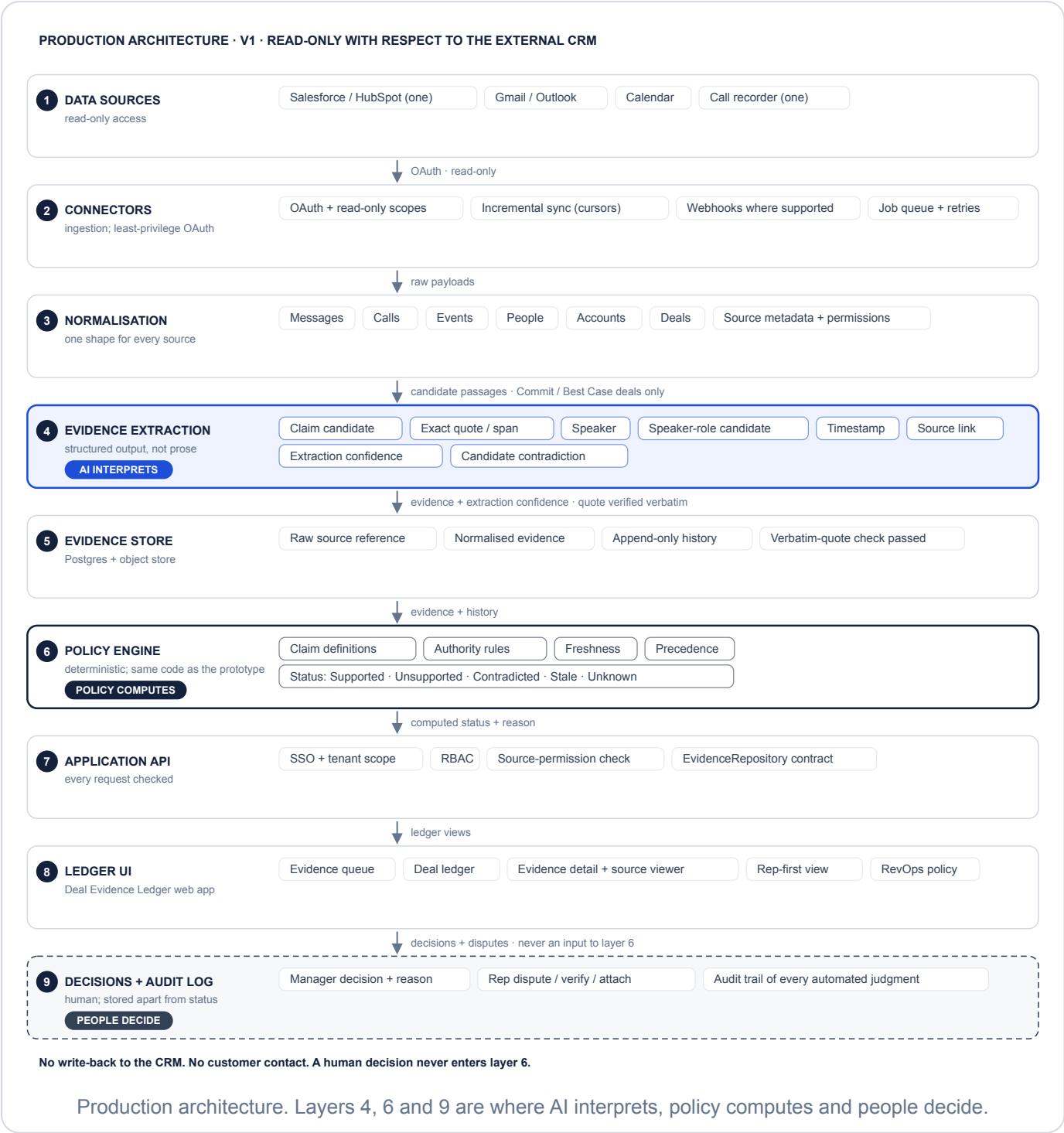
V1 is read-only with respect to the external CRM.

IN	NOT YET (FUTURE POSSIBILITIES ONLY)
Salesforce or HubSpot · email + calendar · one call recorder	CRM replacement · generic AI chat or agent
Commit and Best Case deals · 5–8 predefined claims	Win/loss prediction · autonomous forecasting
Exact quote, speaker, role, time, source link, status, reason, history	Autonomous emails · AI SDR · any customer contact
Rep sees flags before the manager	WhatsApp / chat · buyer portal · relationship graph · renewals
Human decision log, stored separately	Custom rules builder · autonomous CRM write-back

Must prove (proposed thresholds, **not industry benchmarks**): **≥ 90 %** of evidence citations correct and relevant · **> 50 %** of flags judged worth raising by managers · guardrail: reps do not reject it or record less.

Engineering principles. (1) Read-only before write-back. (2) Evidence must always be inspectable. (3) AI proposes interpretation; deterministic policy computes status. (4) Human decisions remain separate from evidence status. (5) Precision before recall. (6) Every automated judgment must be auditable. (7) Tenant isolation and least-privilege access from day one.

2. System architecture



Simplest architecture that carries a pilot: a modular monolith (API, connector, extraction and policy workers), **one Postgres**, one object store, one job queue, and a hosted LLM behind a thin gateway. **Not built:** vector DB, graph DB, Kafka, microservices, agent framework; none is needed for V1, and each has a stated trigger in the appendix. The prototype's **EvidenceRepository** is the Application API contract, and its status engine is the code the policy worker would run.

3. The AI / policy / human boundary, using Castellan

AI INTERPRETS (PROBABILISTIC)	POLICY COMPUTES (DETERMINISTIC)	PEOPLE DECIDE
Finds passages · exact quote · speaker · proposed role · maps to a predefined claim · structures dates and durations · flags candidate conflicts · extraction confidence	Admissibility · authority rank · freshness · precedence · status · evidence review priority · next question	Forecast call · customer contact · missing evidence · disputing a reading · changing the CRM
Never: decides status, predicts win/loss, contacts customers	Never: reads a model score or a human decision	Never: rewrites a computed status

Castellan Freight, *Procurement takes two weeks (CRM says 2 weeks)*. The champion said "about two weeks" on a 22 Sep call. Procurement wrote four weeks on 2 Oct. ① **AI output** for the email (the prototype's exact wire format; the champion passage is the same shape, **0.99**, **CHAMPION**, value **2**):

```
{
  "extraction_id": "ex-cf-proc-email",
  "deal_id": "castellan-freight",
  "claim_definition": "PROCUREMENT_DURATION",
  "source_id": "src-ex-cf-proc-email",
  "source_type": "CUSTOMER_EMAIL",
  "quote": "Our standard procurement review is four weeks after receipt of the complete document set.",
  "speaker": "Hannah Brandt",
  "speaker_role_candidate": "PROCUREMENT",
  "timestamp": "2026-10-02T10:14:00Z",
  "structured_value": 4,
  "extraction_confidence": 0.96,
  "candidate_conflicts": ["ex-cf-proc-champion"],
  "model_version": "extractor-demo-0.1",
  "prompt_version": "claims-v0.1"
}
```

② **Deterministic rule**: accepted authority **Procurement** > **Legal**; fresh for **30 days**; **later or more authoritative evidence wins**. The champion is not an accepted authority, so that passage is context. ③ **Policy output** (real prototype output):

```
{
  "status": "CONTRADICTED",
  "reasonCode": "CONTRADICTED_LATER_OR_HIGHER_AUTHORITY",
  "reason": "Later, more authoritative procurement evidence disagrees: Hannah Brandt (Procurement) states 4 weeks on 2 Oct; the CRM says 2 weeks. Earlier: Marcus Lindqvist (Champion) said 2 weeks on 22 Sep.",
  "controllingEvidenceId": "ex-cf-proc-email",
  "supersededEvidenceIds": ["ex-cf-proc-champion"]
}
```

④ **People decide**: the manager logs *Keep Forecast = Commit* with a reason. The status stays **CONTRADICTED**.

LLM confidence does not determine status. The model was 99 % sure the champion said it. That describes *extraction*. The claim is **CONTRADICTED** because policy found later, more authoritative evidence. A test changes the confidence to 0.01 or 0.99 and the status does not move.

4. AI quality, evaluation and failure handling

Offline evaluation. 50 closed deals × about 8 claims ≈ 400 claim evaluations, labelled by hand by two labellers with adjudication, **point-in-time** (judged as at that week's forecast review, not at close). Dev/test split with a locked test set. Hard cases oversampled: hedged language, conditionals, forwarded threads, multi-speaker calls.

LEVEL	METRICS, REPORTED SEPARATELY
Extraction	Quote/span precision · speaker accuracy · claim-mapping accuracy · role accuracy
Product output	Evidence citation judged relevant and correct · status agreement with a human evaluation under the same policy · flags managers judge worth raising

Threshold [proposed product threshold, not an industry benchmark]: ≥ 90 % citation correctness and relevance. At ~400 evaluations that is about ±3 points overall (±8 within one claim type), so we judge the threshold overall and treat per-claim numbers as directional. **Tune for precision before recall.** If the bar is missed, narrow the claim set first.

FAILURE	SAFEGUARD
Wrong speaker or role	Role shown as <i>proposed by AI</i> on every card; one-step dispute; role accuracy tracked
Hedged remark read as a commitment	Verbatim quote always shown and linked; only accepted-authority customer evidence counts
Channel the system cannot see	"In connected sources" wording; channels shown per deal; rep can add seller-supplied context
Quote out of context	Source viewer shows the surrounding passage; conditionals are in the gold set
Prompt or model change shifts results	Hand-marked set re-runs on every model, prompt or policy change ; a drop blocks release
Prompt injection in a source	Source text is untrusted data; extractor has no tools; schema and verbatim-quote checks

Bounded blast radius: a mistake changes **no CRM record** and reaches **no customer**; the rep can dispute it before the manager sees it.

5. Security, privacy and enterprise readiness

AREA	APPROACH
Access	OAuth per source; minimum scopes, read-only for the CRM ; tokens in a secrets manager with per-tenant envelope encryption; customer admin can revoke at any time
Tenant isolation	<code>tenant_id</code> on every row with Postgres row-level security; per-tenant keys; cross-tenant tests in CI
RBAC	Manager sees their reporting line, rep sees their own deals, RevOps sees policy and aggregates, not individual rep performance
Source-level permissions	A manager is never shown evidence they could not open under company policy. Access metadata is ingested with each source and checked on every evidence read; default-deny when access cannot be established
Consent and recording policy	Only what the company's recorder already records under its own consent settings; we never start recording; private or excluded calls are not ingested
Encryption, retention, deletion	TLS in transit, encryption at rest; we store the quote, a short context window and a pointer, and fetch full text from the source with the viewer's own permission; configurable retention; deletion cascades by source and tenant
Audit	Append-only log of every automated judgment (model, prompt, policy version), evidence view, human decision and policy change
LLM handling	Zero-retention, no-training terms; provider gateway; region pinning if required
Platform dependency	Email, call and CRM APIs can change, be rate-limited or be closed to third parties. Each source sits behind an adapter with contract tests; "sync degraded" is shown, never silently stale; V1 depends on one CRM and one recorder by choice

6. Team, implementation plan, stage gates and top risks

Team (lean): **1 Product Manager** (claim set, evaluation design, gates) · **1 Product Designer** (review experience, rep-first flow) · **2 Full-stack / Backend Engineers** (connectors, evidence store, policy worker, API, UI) · **1 Applied AI / ML Engineer** (extraction, evaluation harness, regression runs). Shared part-time: security / DevOps, QA / data labelling.

WEEKS	BUILD	GATE
0	Design partner and historical audit ; freeze 5–8 claims; first labelled set	G0 value: lost and slipped deals carry more unsupported or contradicted claims than won ones, else stop
1–2	Domain model, CRM connector, email and call ingestion, source viewer, tenant and auth	Real partner data visible in the source viewer
3–4	Extraction pipeline, speaker and claim mapping, evaluation harness	G1 (interim): dev-set precision at an agreed bar [proposed ≈ 85 %] or narrow claims
5–6	Policy engine, queue, deal ledger, rep dispute, decision log	End-to-end on partner data
7	RevOps policy settings, security hardening, audit logs, performance	Pen-test findings addressed
8	Historical backtest and accuracy evaluation	G2 accuracy: ≥ 90 % citations correct and relevant, else narrow and tune
9–10	Design-partner live pilot, rep-first	G3 adoption: > 50 % flags worth raising, disputes low, recording not falling

The historical audit: *This tests whether evidence quality separates outcomes enough to justify a live pilot. It does not establish causality. Do not start write-back until* citation quality has cleared the threshold **and** an adoption signal exists.

TOP PRODUCT/TECHNICAL RISKS

RISK	MITIGATION	WHAT WE MEASURE
1. Authority / role identification	Start with claims where authority is relatively explicit, expose the inferred role, allow one-step correction, measure role accuracy separately, and narrow the claim set if reliability is inadequate.	Role accuracy; disputes citing speaker or role
2. Missing communication channels	Say "in connected sources"; list channels per deal; rep can add seller-supplied context	Disputes citing an unseen channel
3. False extraction / interpretation	Exact quote with context and link; precision-first tuning; regression set on every change	Citation precision, incl. hedged and conditional cases
4. Rep surveillance / adoption	Rep sees flags first; managers see deal evidence, not activity counts; audit reported by pattern, not by rep	Disputes; recording rate (is it falling?)
5. Platform / API dependency	One CRM and one recorder; adapters; contract tests; "sync degraded" status	Sync lag; breaking changes per quarter
6. Evidence quality may not separate deal outcomes	This is the first test , before any live build, judged point-in-time; if it does not separate outcomes, we stop	Unsupported/contradicted rate on won vs lost vs slipped deals

Not every risk is solved; the plan is built so the cheapest tests come first.

Castellan Freight, every account, person and quote is illustrative. Thresholds and policy values are proposals to agree with engineering and a design partner, not industry benchmarks. Generated from docs/ENGINEERING_APPROACH_CONCISE.md.